INFRA-BAYESIANISM FOR PEOPLE WHO DON'T KNOW MEASURE THEORY

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This work was done as part of the SERI MATS John Wentworth-track research sprint (though I'm late getting it out). I collaborated with Lorxus in writing this, who has my thanks for his input. He wrote his own distillation targeting a more mathematical audience, which I link below.

Tl;dr

Infra-Bayesianism is a couple of things - foremost, an alternative theory of probability and a concomittant decision theory. The original sequence is, in my opinion, less understood than it deserves, largely because of the heavy measure-theory / topology background assumed of the reader. My goal in these two(?) posts is to explain Infra-Bayesianism at the level most non-mathematicians know probability. There will be worked examples and exercises - they're pretty hand-holdy, so do them as you see fit; I'll write assuming that you've done them before continuing. For those exercises which involve calculating a handful of things, I made a google spreadsheet for the exercises here, and a separate one for the answers (which I'll link at the end).

You can find the original sequence here, and some not-so-mathy explications here and here. If you want a *more* math-y distillation, see here (disclaimer: Lorxus and I collaborated for our distillations, but decided to write different documents for different audiences). Also just as a general remark, this whole post is a longer, hopefully self-sufficient version of an example given in the post, so all credit for the example selection, and a good portion of my explanations to Diffractor and Vanessa. Ironically, I'm responsible almost solely for the "dumbing down" :)

Disclaimer

There are two parts to this disclaimer: First, the original sequence develops the theory of Infra-Bayesian probability in a way parallel to, say, measure theory in mathematics - the goal of this post is to "dumb it down" enough that high-school probability, plus basic regular, expected-utility maximizing decision-theory should suffice to pull you through. The description I give here is deliberately summarized - it's not meant to give you a feel for how to handle edge-cases, or where the edge-cases are. Apply to anything qualitatively different from these examples at your own risk

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- but IB isn't fundamentally "mathier" than regular probability, so I expect you should mostly be able to get away with what I present for applications.

Second, my background is not in measure-theory; I didn't check the proofs in the original sequence. I've only so far worked through the first 3 basic Infra-Bayesianism posts (and not some of the more "exotic"/recent stuff like PreDCA, naturalized induction), so things I might think are exciting research directions may well have already been solved; cum grano salis. If there are any mistakes here (other than the above-mentioned deliberate over-simplifications), they're my fault. I'll note any material edits directly below here, in addition to changing the post-text.

1. Notation and Decision Theory Recap

(If you're familiar with decision-theory enough to, say, know that π means a "policy", I sugest you skip this section and use the dictionary at the end for notation as needed - much of this will be used only in my second post)

Okay so let's start off by summarizing regular-decision theory, and get set on the notation.

We have an **environment**, e, in which an agent operates according to their **policy**, π, (our goal is to set the best policy for a given environment). We'll be considering discrete environments, which, at each timestep, give some **observation**, $o \in \mathcal{O}$ to the agent (sometimes the observation is drawn from a probability distribution, and this distribution is allowed to depend on the 'hidden-state' of the environment - then the agent has to do some fancy inference). The agent must then select an **action**, $a \in \mathcal{A}$. At each time-step, the agent gets a **reward** $r(s, a) \in \mathbb{R}$. We'll consider the case where an agent has a deterministic policy (as this is the scenario IB plays in, contending that we should consider any source of randomness as part of the stochastic environment in which the agent is embedded). We'll also ignore the 'hidden-state'; that is,

$$\pi : (\mathcal{O} \times \mathcal{A})^n \to \mathcal{A} \tag{1}$$

$$e : (\mathcal{O} \times \mathcal{A})^n \times \mathcal{A} \to \Delta\mathcal{O} \tag{2}$$

A policy takes any "history" (some list of all the past observations + actions the agent has gotten/done) to an action. The environment takes histories to probability-distributions over observations.

The name of the game is to maximize the agent's expected reward; calling $e(\pi)$ the overall probability distribution of the environment when the agent plays π, we want to find

$$\pi \in \text{argmax}_{\pi^*} \mathbb{E}_{e(\pi^*)} \left[\sum_t \gamma^t R_t(\tau) \right] \tag{3}$$

(capital letters inside expectations are random variables, lower-case letters are instantiations of the corresponding variable). Note that we abbreviate "histories" with the letter τ (which stands for 'trajectories'... eh.)

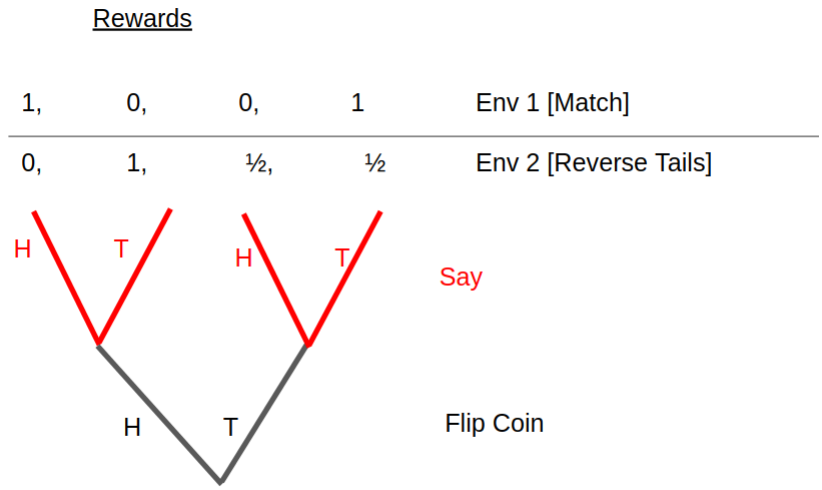

FIGURE 1. Game Tree for the two environments we consider - "Match" and "Reverse Tails". The gray lines indicate a random transition (50-50 odds) - you can see the outcome of this. Afterwards, you say either "Heads" or "Tails" - the reward you get is shown for each environment above that outcome.

EXAMPLE 0: GOOD-OLD DECISION THEORY AND ENVIRONMENT UNCERTAINTY

(Even if you understand decision-theory, please look at the game-tree figure, since this is the example we'll be using)

Let's describe a game: (as in figure 1 I flip a coin and show it to you; you then choose to say either "Heads" or "Tails". Now the thing is, you're uncertain what the rules of the game are - you don't know what the payouts are for each observation-action pair. You give $p = 0.3$ that you're playing game 1 ("Match"), in which you win 1\$ if you say what the coin shows, else 0\$, and $p = 0.7$ that you're playing game 2 ("Reverse-Tails"), in which you get 0.5\$ if the coin shows tails, no matter what you do; if the coin shows heads, 0\$ if you match it, 1\$ if not.

Policies are of the form "if I see H, then X; if I see T, then Y", for X and Y your actions - we'll abbreviate policies as "XY".

EXERCISE 0: Work out the expected reward for each of the four policies - do it for each environment individually, and then find the overall EV supposing you had a prior for, say, 30% of being in "Match" vs. 70% of being in "Reverse Tails". If it's any help, you can fill out the table below:

Hopefully this is a good refresher on how Bayesian-expected-utility-maximizers normally handle environment-uncertainty. Get ready to contrast this with the way IB handles it.

(Abbreviate our policies “if see H, do X, if see T, do Y” as “XY”)

Policy (action if see H, action if see T)	EV in env 1	EV in env 2	EV for 30-70 prior over envs 1 and 2
HH			
HT			
TH			
TT			

	Env 1	Env 2	30-70 Prior
Optimal Policy			
EV of Optimal Policy			

FIGURE 2. Table for exercise 0; you can find an google-sheets version in the exercise sheet here

EXAMPLE 1: KNIGHTIAN UNCERTAINTY

Infra-Bayes and Knightian Uncertainty. Okay so one of the big differences between Infra-Bayes and Normal-Bayes is that we want a coherent idea of Knightian uncertainty. Knightian uncertainty is when ”you’re so uncertain between such-and-such things that you daren’t assign them a probability”. Now for context, normally all the ”money-pump” arguments indicate that Knightian uncertainty is for chumps and/or cowards - you should just slap a probability on it on pain of being Dutch-booked.

Thing is, in IB, Knightian uncertainty isn’t supposed to model your uncertainty about your environment (regular Bayes handles this just fine) - instead, it asks ”what would decision theory look like if I divided up possible-universes into equivalence classes, which are distinguishable between, but not within classes?”. The central example of concern is where an agent *literally doesn’t have the representation power to represent these two universes as distinct* - e.g., if it is modeling itself (modeling itself modeling ...) + the universe. Obviously there’s some missing architecture for how an embedded agent **does** Infra-Bayesianism (computationally) if it, like, can’t by hypothesis, but, eh, ”future research direction”. Keep that in mind.

Now, one more ingredient for Infra-Bayes, and this is IMPORTANT and also SOMEWHAT ARBITRARY - we suppose that, within one of these equivalence classes of indistinguishable-universes

we'll assume that, contra-Leibnitz, we're in the worst possible one. Btw, I'm gonna start calling one such equivalence class a "bucket" of universes - from now on, unless I say otherwise, we'll assume that everything is together in one bucket. We're being conservative here - you could plausibly argue for some other decision rule, but maximin is as good a choice as any, and the math works out nicely.

A lot of the weird-feeling stuff in IB (e.g. dependence on counterfactuals) comes from this assumption, so take it as you will. It also is worth noting that choosing the best hypothesis in IB is equivalent to doing maximin on a sequential zero-sum game - your "action" is your *whole policy* and your opponent's action is choosing which environment e you get put in within a "bucket" (after they see your policy choice). The payout is your ex ante expected utility from playing X policy in Y environment. This is only true within a single "bucket" - having more just requires another regular prior over which bucket you're in. It is customary to refer to this opponent as "Murphy" (of law fame) or "the Demiurge" (of Gnostic fame). I tend to disprefer these, but if it helps you, feel free to think about it that way.

As a side note, I've seen it said here that a broader equivalent class of "Nash-bargaining" problems can still be put in the IB framework - I haven't read this in any detail yet.

Example 1. Okay onto the example: same game as above, but now, the universes of game 1 and game 2 are indistinguishable to you. They're in the same "bucket". To be extra clear - the state, observation, and action spaces are the same in these two environments - they differ only in parameters like reward and transition-probabilites. You select some complete (i.e. prescribing an action for every possible eventuality) policy at the start, and that immediately sets what environment you're playing in. Whatever your policy is, calculate its EV in each environment, then take the lesser of those, and call *that* the "infra-expected value over this bucket of universes" (this is just motivation - we'll talk more about it below)

Again, for the game tree, see Figure 1 above.

To make really clear how "Knightian uncertainty" changes thing, here's an exercise:

EXERCISE 1: Figure out what the best policy is (N.B. it might be a little confusing where the "worst-possible world" assumption comes in - that's okay - highlight the confusion and look at the hints!)

(Here more than elsewhere I'm going to assume you did/understood that exercise before continuing) If you grokked this or the exercise earlier, you should be weirded out. Think about what I mean for a minute, then let's highlight why, just-in-case:

So *ex ante*, your best strategy is to play double-tails; it's your best only because your policy doing well on one world branch effects the outcomes you should expect in the other one - whacky! I.e., playing TT gets you the reward from environment 2, and in environment 2, TH should get you the same reward, but actually trying to play TH will get you booted out of env 2 into env 1!

Worse yet, suppose I just showed you my coin came up H - then, if you were being dogmatic about "Knightian uncertainty", you're facing down a (1,0) and (0,1) payout - and you're thus indifferent between saying heads or tails. You thought ahead and planned exactly what you would do on seeing

(Abbreviate our policies “if see H, do X, if see T, do Y” as “XY”)

Step 1. Calculate EV of each policy in each environment in the “bucket”

Policy (action if see H, action if see T)	EV in env 1	EV in env 2	Worst environment for this policy	Reward in worst environment
HH				
HT				
TH				
TT				

Step 2. The “infra-EV” of given policy is the minimum-over-environments of EV[pi]

Optimal policy	
(Infra) EV of optimal policy	

Step 3. Choose policy with the best infra-EV (the policy with the best reward when put in its own worst environment)

FIGURE 3. Table for exercise 1; you can find an google-sheets version in the exercise sheet here

H ex ante, and as soon as you do, you go ”nope” and do something else. You predictably disagree with yourself, and your decision theory just doesn’t work - we need to be smarter about this wheole ”Knightian uncertainty” uncertainty business.

EXERCISE 1.1: What do you need to change about your update-rule in order to get you-after-seeing-heads to agree with you-before-seeing-an-observation? (this is a thinker - keep reading for the answer, but it won’t be in the google-sheet)

The following paragraph has some hints to deconfuse you, but try and get confused first. This is relatively easy to grok if you work through it but not easy to say in words. The answer is broken up into a series of hints below, after each of which I invite you to try your hand again at the exercise before reading the succeeding one :)

(BEGIN HINT 0)

We’re committed to this ”worst of all possible worlds” thing, but we specifically mean ”worst for your *ex ante* expected utility given your policy”, not ”lose and cry - every random event is optimized against you lol”. As soon as you set your policy at the start, that sets what the worst possible world for that policy is (in ex-ante expected utility), and then everything plays out according to that particular environment’s rewards and probabilities. BUT this means that, when you update

on an observation, what you *would have done counterfactual to that observation* effects what world-branch you're in, and thus what you should do. That's because your counterfactual self getting a ton of reward in, say, environment e, will make it hard for e to be the worst ex ante. You're not in Kansas anymore, and you can't just do normal Bayesian updates.

(END HINT 0)

(BEGIN HINT 1)

Given that you're in the "Heads" branch, what about the counterfactual branch do you need to know in order to make sure your updated judgement is calculated the same way as your *ex ante* judgement? The answer should be a scalar. Assume that *ex ante* you doesn't need to change anything - just change what post-observation-you does.

(END HINT 1)

EXERCISE 1.1 ANSWER:
Okay so the problem has the form max-over-policies AFTER inf-over-environments OF reward-based-on-policy-in-environment. Basically, you're "integrating" out the counterfactual policy and utility - the expected utility in the counterfactual is the scalar you want - then you do max-over-post-observation-policies AFTER inf-over-environments of [(EV of post-obs-policy given the observation) PLUS (EV of counterfactual policy given counterfactual observation)]. That last-term is the scalar we're gonna start lugging around.

So really our "Infra-distribution" is more like a handful (one for each environment) of 2-tuples of the form ("function which eats on-history policies and spits out utility", "scalar of running counterfactual EV, given for some specific off-history policy"). If you're looking to follow the original sequence, this is the simplified version of what they call an "a-measure".

Recap and Some Quick Remarks. Alright so we found out that we need to do Bayesian updating but also keep track of what the off-history reward was. Note two things: first, so far, we only figured out "updates" for when you already set your full policy in advance (which is weird), and second, note that the utility function plays a priveleged role in deciding what world we get put in.

By "priveleged" role, I mean that, in normal decision theory, there's a really clean line between "probability thing X happens" and "value of reward (or whatever) when X happens" - one is the thing you *use* to take expectations, and the other is the thing you *take expectations of*. Now we have some kind of coupling between "probability" and "value" that we'll spend the next while trying to make really clear.

Interlude on Infra-Bayesian Probability Theory

(The goal of this section is to introduce notation for the stuff we just did (which you should understand), and notation for some stuff we're about to do (which will be explained in the next examples - I just wanted to have all this in one place).

FORESHADOWING / NOTATION NOTE: We assume all reward is in the closed interval [0,1] - this is equivalent to assuming the reward is bounded at all, and is just rescaled.

Okay! If you feel so inclined, take a minute and try and answer the question/vibe "wait so why do we need counterfactuals? That seems like a really weird thing to include in decision theory, and this doesn't *seem* like we're doing UDT yet?" (I talked about this earlier, but it's good to re-think it as we go).

So in the language of earlier, here we're going to be talking all within one "bucket" of indistinguishable universes in this part ("we have a single infra-distribution H" - more on this in a moment). We're going to sidestep a *lot* of the math from the original sequence by assuming - for no reason other than making things way simpler - that this bucket has only finitely many universes in it. Now bear with me while I write a bunch of symbols, and I'll explain them right after:

An infra-distribution is basically a set of tuples $(\mathbb{P}_e, c_e, \lambda_e)$ (for each environment e) We'll generally use the letter H for these infra-distributions. Specifically an infra-distribution looks like:

$$H = \{(\mathbb{P}_{e_i}, c_{e_i}, \lambda_{e_i})\}_{i \leq N} \tag{4}$$

(N is just however many environments are in the bucket - if you want infinitely many environments, well uhhh go learn measure theory and read the original sequence :P)

Also, like a probability-distribution, we require it be normalized to one - I'll explain how you "normalize" these tuples in a moment.

What are these tuple entries? First and simplest, $\mathbb{P}_e$ is our old friend, "regular probability" of some outcome occuring given we're in environment e. If you like the specificity, you can assume it's a probability distribution over possible partial action-observation histories (i.e., sequences $(o_1, a_1, ...o_t, a_t)$, where t may or may not be the final timestep). We'll just assume that whenever we want to ask about some probability in environment e, $\mathbb{P}_e$ can answer it.

Second, c (a scalar) is where we store information about our off-history reward. We'll be more specific about exactly how we encode this in a bit.

Last, λ is a scalar representing something like "plausibility that we're actually in this environment e" (NOT "probability" though - again, we'll get there in a minute)

So this whole "what's the *ex ante* worst-case outcome in this bucket for a particular policy?" question we formalize by making it the new "infra-expectation value". We're about to start prefixing a lot of stuff with "infra", and I'll use tildes to indicate "infra- version of traditional probability-theory concept", like $\mathbb{E} \to \widetilde{\mathbb{E}}$ but this is more of a training-wheels thing, as no one else does as far as I'm aware.

Okay! FIRST a notational note - we're going to start blurring the lines between utility and probability, so we will assume rewards are all bounded in the closed interval $[0, 1]$ (which is equivalent to just assuming they're bounded at all. NOW, three out-of-the-blue definitions, then an explanation:

First, an infra-expectation value

$$\widetilde{\mathbb{E}}_H[U] = \inf_{(\mathbb{P}_{e_i}, c_{e_i}, \lambda_{e_i}) \in H} \lambda_{e_i} \left(\mathbb{E}_{e_i}[U] + c_{e_i} \right) \tag{5}$$

Second, a gluing operator (just because the shorthand is nice given we'll have to use it a lot - there's nothing "infra-" about it):

$$(f \oplus_h g)(x) = \begin{cases} f(x), & \text{where } h \text{ happens} \\ g(x), & \text{where } \neg h \text{ happens} \end{cases} \tag{6}$$

Third, infra-probability

$$\widetilde{\mathbb{P}}_H = \widetilde{\mathbb{E}}_H[1 \oplus_h U] - \widetilde{\mathbb{E}}_H[0 \oplus_h U] \tag{7}$$

(the 1 and 0 represent maximum and minimum reward btw)

Explanation time: First, the infra-expectation-value $\widetilde{\mathbb{E}}_H[U]$ means "what's the expected value of playing some fixed policy, in the *worst* environment for that policy" - the λ and c bits handle some scaling. We'll talk about updates momentarily, but you can think of it as "the worst expectation" but affine-transformed in some way so that if I ask after doing an update, I agree with myself ex ante.

Second, the gluing operator $\oplus_h$ is just a glorified/faster "if" statement. We'll usually have the first operand be "utility on-history" and the second "utility off-history", with the h in the $\oplus_h$ the history we actually see.

The infra-probability $\widetilde{\mathbb{P}}_H(h)$... is a bit hard (i.e. best to understand by doing some examples). It means something like "value-at-stake, given that you've seen some history h". For instance, for the $\widetilde{\mathbb{E}}_H[1 \oplus_h U]$, it's probably best to think of asking yourself, ex ante, "What would be my expected utility if I got maximum reward after h no matter what happened next, and normal reward otherwise?" (1 is the "max reward", remember). Likewise the $\widetilde{\mathbb{E}}_H[0 \oplus_h U]$ is the same, but replace "maximum" with "minimum". In normal decision-theory this is trivial because the off-history bits cancel out.

In IB, they might not, if getting a lot of reward vs. little on h would mean you change what environment is your policy's worst. It's a sort of counterfactual question - "what would happen if the reward structure was Y instead of X, vs. if it was Z". Note, too, that it's totally funky to treat this as a probability - it would have units of *dollars* (or utils) if we hadn't scaled it already to be in $[0, 1]$! But we will ¿:D The reason we call this a "probability", btw, is that it will be in all the same spots as normal probability would when we go to write down infra-analogues of stuff like Bayes' rule - if you don't like it, you can call it something else just as well; it doesn't actually obey probability/Kolmogorov axioms.

Also, just to be really clear: INFRA-EXPECTATIONS ARE NOT (in general) LINEAR!! Specifically, they're the infimum of a bunch of affine things, which is concave: $\widetilde{\mathbb{E}}_H[A + B] \geq \widetilde{\mathbb{E}}_H[A] + \widetilde{\mathbb{E}}_H[B]$.

EXERCISE 2: Calculate $\widetilde{\mathbb{E}}_H[U]$, $\widetilde{\mathbb{E}}_H[0 \oplus_h U]$, and $\widetilde{\mathbb{E}}_H[1 \oplus_h U]$ (U is the utility/reward) for each policy in example 1 (i.e. the same game tree we've been using the whole time), and $\widetilde{\mathbb{P}}_H(o)$ for each observation and each policy.

Policy (action if see H, action if see T)	Infra-EV overall	Infra-EV of [0 +_(Tails) U]	Infra-EV of [1 +_(Tails) U]	Infra-prob of seeing tails
HH				
HT				
TH				
TT				

Policy (action if see H, action if see T)	Infra-EV of [0 +_(Heads) U]	Infra-EV of [1 +_(Heads) U]	Infra-prob of seeing Heads
HH			
HT			
TH			
TT			

FIGURE 4. Table for exercise 2; you can find an google-sheets version in the exercise sheet here

Remember that it's okay to be a little weirded out and confused - hopefully once you see the update rule, you'll reach maximum confusion, but then have everything you need to work through some deconfusing examples.

That said, let's get to it: let's about the formal "update rule" for infra-distributions. Remember that so far, we've only talked about cases where your policy is fully-chosen from the start - that means you're not really "updating" in the sense of changing your beliefs in a way that effects your actions. Call the "updated" distribution $H|_h$, for h the history we're updating on and H the ex ante infra-distribution.

I find it most useful to think of $H|_h$ as an algorithm whose job is to take in questions "from the perspective of h" (functions only defined on histories which have h as their prefix), and answer them the same way H would if you gave it $(\cdot) \oplus_h U$. I.e., let f be our utility-function on h, and g the off-h utility; then $\widetilde{\mathbb{E}}_{H|_h}[f]$ "isn't equal to but is closely related to" $\widetilde{\mathbb{E}}_H[f \oplus_h g]$. I'll give the nice relationship after we define it below.

Okay, the now for the rule itself: remember, we can get everything we want from an infradistribution by taking expectations for different functions; when we make an observation h (and really, again, it's better to imagine your ex ante-self asking "if I were to observe h..."), we change infradistributions $H \to H|_h$.

This updated infradistribution is defined as:

$$\widetilde{\mathbb{E}}_{H|_h}[U|_h] = \frac{\widetilde{\mathbb{E}}_H[U] - \widetilde{\mathbb{E}}_H[0 \oplus_h U|_{\neg h}]}{\widetilde{\mathbb{E}}_H[1 \oplus_h U|_{\neg h}] - \widetilde{\mathbb{E}}_H[0 \oplus_h U|_{\neg h}]} \tag{8}$$

Here I wanted to be really explicit; $\neg h$ is "not h". I also marked U as being restricted to h or $\neg h$ (by writing $U|_h$ and $U|^{\neg h}$ - the bar means "restricted to" in this context, but "update based on" for $H|_h$). If you're willing to let it be implied, (as I usually am),

$$\widetilde{\mathbb{E}}_{H|_h}[U] = \frac{\widetilde{\mathbb{E}}_H[U] - \widetilde{\mathbb{E}}_H[0 \oplus_h U]}{\widetilde{\mathbb{E}}_H[1 \oplus_h U] - \widetilde{\mathbb{E}}_H[0 \oplus_h U]} \tag{9}$$

This is pretty and simple. I'll give you an uglier but maybe more intuition-accessible version below, but first:

EXERCISE 2.1: Prove that $\widetilde{\mathbb{E}}_H[f \oplus_h g] = \widetilde{\mathbb{E}}_H[0 \oplus_h g] + \widetilde{\mathbb{P}}_H(h)\widetilde{\mathbb{E}}_{H|_h}[f]$. Write out what this means in words. (This is another thinker, no answer in the google-sheet)

EXERCISE 3: For the two-environment game set up in Example 1, calculate, for each policy, and for each observation, the quantity $E_{H|_o}[U]$ (using the formula above).

Policy (action if see H, action if see T)	Infra-EV of H|_(Tails)	Infra-EV of H|_(Heads)
HH		
HT		
TH		
TT		

FIGURE 5. Table for exercise 3; you can find an google-sheets version in the exercise sheet here

Also note at this point that we're really only talking about expectations over utility-functions (maybe with a piece papered over with 0 or 1). IB has machinery to do arbitrary expectations, but you just have to add a lot more symbols and I figured I'd keep it simple.

It's nice that that defines the infra-distribution update, but we're probably more comfortable talking in terms of the tuples that remind us a little more of probability distributions.

If you want to update a tuple, we have a nastier but maybe more intuitive rule:

$$(\mathbb{P}_e,\ c_e,\ \lambda_e) \rightarrow_{\text{see } h} \left(\mathbb{P}_e|_h,\ \frac{c_e + \mathbb{E}_e\left[0 \oplus_h U\right] - \frac{1}{\lambda_e}\widetilde{\mathbb{E}}_H\left[0 \oplus_h U\right]}{\mathbb{P}_e(h)},\ \lambda_e \frac{\mathbb{P}_e(h)}{\widetilde{\mathbb{P}}_H(h)} \right) \tag{10}$$

EXERCISE 4: Update the tuples from Example 1, for each observation (λ starts as 1 and c as 0 at the beginning of the game)

Policy (action if see H, action if see T)	C in env 1, given you see Tails	Lambda in env 1, given you see Tails	C in env 2, given you see Tails	Lambda in env 2, given you see Tails
HH				
HT				
TH				
TT				

Policy (action if see H, action if see T)	C in env 1, given you see Heads	Lambda in env 1, given you see Heads	C in env 2, given you see Heads	Lambda in env 2, given you see Heads
HH				
HT				
TH				
TT				

FIGURE 6. Table for exercise 4; you can find an google-sheets version in the exercise sheet here. And yes, the format of the table is itself a hint ;)

EXAMPLE 2: FOR A FEW PRIORS MORE

So hopefully you've worked through the last bit and have a good feel for what's going on. You may be a little weirded out by the "Knightian uncertainty + pessimism = counterfactuals matter now" thing, and hopefully this example will weird you out even more. In this section I'm going to introduce _two_ new things. New-thing-1 is having a normal prior over infra-distributions (this is pretty uncomplicated tbh). New-thing-2 is figuring out what happens to how "Knightian uncertainty" handles evidence for or against being in a given one of these supposedly indistinguishable worlds.

1.1. **New-Thing-1: Priors Over Infradistributions (or "Mixtures").** First, the prior over infra-distributions is dead simple: suppose we have a handful of infradistributions H_i, and a prior over them ζ_i. ζ is just a totally normal (here, discrete) probability distribution just like it would be if you were indexing dice outcomes - ζ_i are just good-old-fashioned probabilities, like 0.3.

This is very much like "I think 40% I'm in water-is-wet world, and 60% I'm in water-is-dry-world - I can poke water and get evidence about which one. But I can't readily store hypotheses about the molecular configuration of my glass of water in either case - this configuration definitely is related to the macroscopic wet/dry-ness, and vice versa,though!". The 40-60 split is ζ here, and the molecular configurations are in H_{dry} and H_{wet}, respectively.

So this prior over infradistributions turns out to be equivalent to just having one big infradistribution (which isn't surprising, since it's basically the same in regular decision theory). Let's call this new "mixture" infra-distribution by the name $\mathbb{E}_\zeta[H_i]$ - this is a *new* infra-distribution - don't let the regular-looking expectation-value fool you. It gives us a new pile of environments-cum-bookkeeping-scalars. As before, it's easier to talk about calculating expectations with this distribution first, and then talk about what the individual tuples look like.

All we need to know to define infra-distributions is to calculate expectation-values over arbitrary functions, and for priors over them, that works out like:

$$\widetilde{\mathbb{E}}_{\mathbb{E}_\zeta[H_i]}[f] = \mathbb{E}_\zeta\left[\widetilde{\mathbb{E}}_{H_i}[f]\right] = \sum_{1\leq i\leq N} \zeta_i \widetilde{\mathbb{E}}_{H_i}[f] \tag{11}$$

Easy as pie - you just add 'em up. What do the tuples in the mixture distribution look like? Weighted sums of the tuples in H_i. More specifically, suppose I had H_1 with env 1a and env 1b, and also H_2 with env 2a and env 2b. Let's say I have $\zeta_1 = 0.9 = 1 - \zeta_2$. Then the mixture has 4 tuples in it: first, 0.9*(1a) + 0.1*(2a), second, 0.9*(1a) + 0.1*(2b), third, 0.9*(1b) + 0.1*(21), and fourht, 0.9*(1b) + 0.1*(2b).

By the way, if you're confused how we're doing algebra with these tuples, it's actually pretty simple; let a and b be two scalars, then

$$a\left(\mathbb{P}_1,\ c_1,\ \lambda_1\right) + b\left(\mathbb{P}_2,\ c_2,\ \lambda_2\right) = \left(a\mathbb{P}_1 + b\mathbb{P}_2,\ ac_1 + bc_2,\ a\lambda_1 + b\lambda_2\right) \tag{12}$$

Much less arcane than it might sound.

Okay so that's new-thing-1 - I'm gonna introduce new-thing-2 now and only *then* will there be an exercise.

1.2. **New-Thing-2: Evidence Within Buckets.** Let's change the game a little bit - everything is the same except the coin is no longer fair. In environment 1, it's heads-weighted, and in environment 2, tails-weighted (say prob = 0.9 vs 0.1 in each case, m.m.). This is going to wind up a sort of attack on the idea of "Infra-Bayesian probability as a theory of Knightian uncertainty", because now, as soon as you make an observation you have a lot of evidence about which of two supposedly indistinguishable environments you're in, and we're not just gonna throw good evidence away.

Tl;dr is that observations which give you evidence *are* allowed to change your credence about what environment you're in - except now it's a little weird and value-weighted-y.

This was actually already built into the formulas I gave you for updating during the interlude. Take a look at equation 10. It's all in the $\mathbb{P}_e(h)$ - the third entry, the λ one, is related to how much we believe we're really in environment e - it almost looks like it gets updated by a likelihood ratio! I expect this to make sense if and approximately-only-if you do the upcoming exercise.

Now you should be ready for the coup de grace: Infra-Bayes' rule!!! Infra-Bayes rule is how you update a mixture of infra-distributions, to get something like $(\mathbb{E}_\zeta [H_i])|_h$ (Remember! This outputs a new infra-distribution, not a number or anything!)

EXERCISE 4.2: Prove Infra-Bayes' rule:

$$(\mathbb{E}_\zeta [H_i])|_h = \frac{\mathbb{E}_\zeta \left[\widetilde{\mathbb{P}}_{H_i}(h) H_i|_h\right]}{\mathbb{E}_\zeta \left[\widetilde{\mathbb{P}}_{H_i}(h)\right]} \tag{13}$$

Okay now to cap things off with a big exercise that, if it's not tedious now, it hopefully will be by the end of it.

EXERCISE 5 / HOMEWORK: We have two infradistributions H_1 and H_2. See the game tree below for all relevant parameters. Calculate everything. (By which I mean everything in the google-sheets table here, and just until you feel like it's stopped being worth your time)

This "homework" sheet is quite large (though all the calculations are very simple and very similar to the ones you've done so far) - you are welcome to write a program to do it for you if you think it would be faster.

Dictionary

The main difference between my treatment here and the original sequence is dropping as much as possible of the measure-theory. A measure is a probability-distribution, but with a scale factor. The original sequence deals in "signed-affine measures" (where the probabilities can be negative within some restrictions) - but requires that an infradistribution have "positive minimals". The minimals determine the behavior of the infra-distribution, so we'll just deal with them. Since the minimals are "positive" (have no negative probability), we can just keep track of a probability distribution and its scale-factor.

Specifically, given an affine-measure $(m, b) = (\lambda\mu, b)$, this translates into our $(\mathbb{P}, c, \lambda)$ notation as $\mathbb{P} = \mu$, $c = b/\lambda$ and $\lambda = \lambda$.

In our treatment, we consider only updates with respect to a fixed utility function U - so what the original sequence dubs as $H|_h^g$, we assume the off-history utility g is always $U|^{\neg h}$, and leave it implicit.

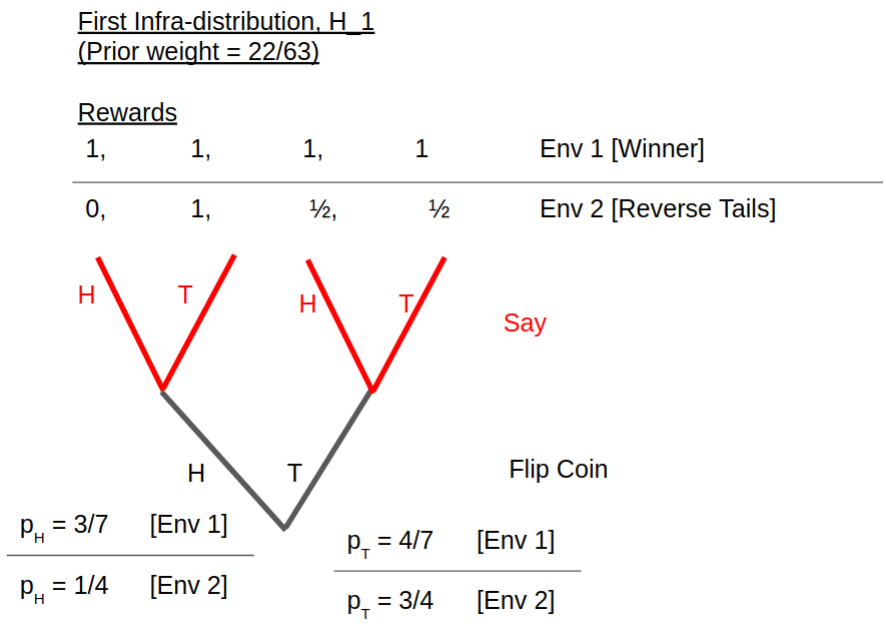

FIGURE 7. Game tree for exercise 5, for the first infra-distribution H_1.

Additionally, the gluing operator we use $\oplus_h$, where the original sequence uses $\star^h$. No excuse here save for personal preference and syncing with Lorxus' distill :)

KEY POINTS

Here's a summary of the things you should have learned from this part of the tutorial

- IB is sorta-kinda a theory of Knightian uncertainty - its more meaningful to say its a probability / decision theory for when you have *non-realizable hypotheses*.
- The central object of IB is the "infra-distribution. An infra-distribution is, roughly, a set of environments, with some extra scalars for bookkeeping. It's the infra-analogue of a probability distribution.
- IB decision theory is pessimistic over an infra-distribution: once a policy is selected, we assume that we are put in the environment which is the worst (in terms of ex ante expected-value) for that policy.
- IB policy-evaluation depends on counterfactuals - and this is due to the ex ante pessimism assumption + the dynamic consistency requirement (NOT to any weird Newcomb-problem-y blahblahblah - it's part of the theory, not of a particular problem).
- The utility function plays a priveleged role in IB - value and probability are (anti-)correlated.

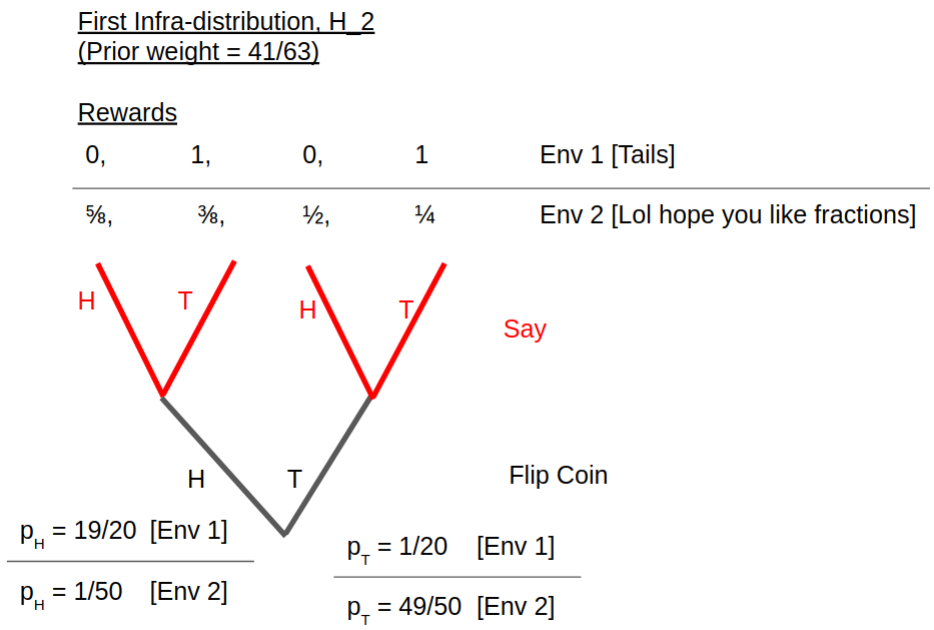

FIGURE 8. Game tree for exercise 5, for the second infra-distribution H_2.

- IB has analogues for many concepts in probability theory - infra-expectation value, and infra-probability in particular. You should have learned how to compute these, given an infra-distribution.
- You should have learned how to update an infra-distribution in two ways: first, by defining how to calculate all possible infra-expectations, and second, from updating each individual tuple-element of the infra-distribution.
- "Mixtures" of infra-distributions are like having a prior over infradistributions (just like in regular learning theory, one has a prior over possible environments).
- We can update (mixtures of) infra-distributions based on observations with infra-Bayes' rule.

ANSWERS TO EXERCISES

See here for the exercise sheet (blank) See here for the answers sheet (i.e., filled in)